

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12404064
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Yu; Haijun et al.

Fixing assembly and fixing device

Abstract

A fixing assembly includes a connecting sleeve, a spike member, and an elastic member; the spike member is rotatably connected to the connecting sleeve; one end of the elastic member is connected to the connecting sleeve, and the other end of the elastic member is connected to the spike member; the elastic member at least helically surrounds and abuts against the outer peripheral wall of the spike member, and the elastic member is configured to generate elastic deformation when the spike member rotates relative to the connecting sleeve along a direction opposite to the helical surrounding direction of the elastic member; there is an included angle between the direction of elastic deformation and the axial extension direction of the spike member to form a helical convex elastic structure.

Inventors:	Yu; Haijun (Guangdong, CN), Chen; Kang (Guangdong, CN), Xie; Yinghao (Guangdong, CN), Zhang; Xuemei (Guangdong, CN), Li; Aixia (Guangdong, CN), Li; Changdong (Guangdong, CN)
Applicant:	GUANGDONG BRUNP RECYCLING TECHNOLOGY CO., LTD. (Guangdong, CN); HUNAN BRUNP RECYCLING TECHNOLOGY CO., LTD. (Hunan, CN); HUNAN BRUNP EV RECYCLING CO., LTD. (Hunan, CN)
Family ID:	78961069
Assignee:	GUANGDONG BRUNP RECYCLING TECHNOLOGY CO., LTD. (Foshan, CN); HUNAN BRUNP RECYCLING TECHNOLOGY CO., LTD. (Changsha, CN); HUNAN BRUNP EV RECYCLING CO., LTD. (Changsha, CN)
Appl. No.:	18/283416
Filed (or PCT Filed):	May 27, 2022
PCT No.:	PCT/CN2022/095690

PCT Pub. No.: WO2023/020064

PCT Pub. Date: February 23, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240174400 A1	May. 30, 2024

Foreign Application Priority Data

CN	202110961094.9	Aug. 20, 2021
----	----------------	---------------

Publication Classification

Int. Cl.: B65B69/00 (20060101)

U.S. Cl.:

CPC B65B69/0075 (20130101);

Field of Classification Search

CPC: B69B (69/0075); F16B (21/086); F16B (21/02); F16B (21/06); F16B (21/065); F16B (21/125); F16B (37/041); F16B (37/042); F16B (5/0266); F16B (2019/006); F16B (13/0875); F16B (13/0841); F16B (13/0833); F16B (13/061); F16B (13/003); F16B (13/04); F16B (29/00); B65D (88/54); B65D (2231/001); A61B (17/869)

USPC: 411/15-80; 411/347; 411/438

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1788270	12/1930	Baranoff	N/A	N/A
2913953	12/1958	Tendler	29/524.1	F16B 13/061
2914983	12/1958	Kopf	N/A	N/A
3373647	12/1967	Sherock	411/16	F16B 37/12
3381566	12/1967	Roy	411/36	F16B 13/061
3881393	12/1974	Campbell	411/16	F16B 13/061
4474515	12/1983	Pitzer	411/16	F16B 29/00
4722649	12/1987	Ferguson	24/453	F16B 21/00
4812097	12/1988	Moulton	411/347	F16B 21/02
4832548	12/1988	Strobel	411/42	F16B 19/1063
2013/0226251	12/2012	Chegini et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2043687	12/1988	CN	N/A
2384033	12/1999	CN	N/A
103407797	12/2012	CN	N/A
204606707	12/2014	CN	N/A
113002906	12/2020	CN	N/A
113262030	12/2020	CN	N/A
113830445	12/2020	CN	N/A
1183249	12/1969	GB	N/A
2137157	12/1983	GB	B65B 69/0008

OTHER PUBLICATIONS

International Search Report for PCT/CN2022/095690 mailed Jul. 29, 2022, ISA/CN. cited by applicant

Primary Examiner: Impink; Mollie

Attorney, Agent or Firm: Westbridge IP LLC

Background/Summary

(1) The present application is a National Phase entry of PCT Application No. PCT/CN2022/095690, filed on May 27, 2022, which claims priority to Chinese Patent Application No. 202110961094.9, titled “FIXING ASSEMBLY AND FIXING DEVICE”, filed on Aug. 20, 2021 with the China National Intellectual Property Administration, which are incorporated herein by reference in their entireties.

FIELD

(2) The present application relates to the technical field of new energy material recycling, and in particular to a fixing assembly and a fixing device.

BACKGROUND

(3) A waste battery, after being recycled, needs to be disassembled, dried, pyrolyzed, crushed and screened, and then acid-leached with sulfuric acid to remove impurities such as copper, iron, and aluminum in the battery powder before entering subsequent processes. At present, the battery powder is generally packaged and transported in ton bags. After the ton bags are transported to the designated places through turnover boxes, the bottom of each ton bag should be simply fixed, so as to prevent the ton bag from falling into a work bin when it is dumped. However, when the ton bag needs to be fixed, simply the four corners or edges of the upper end of the ton bag can typically be fixed to facilitate the ton bag dumping because the ton bag is heavy and the space around the turnover box is compact, which makes the ton bag difficult to locate.

SUMMARY

(4) An object of the present application is to overcome the shortcomings in the conventional technology and provide a fixing assembly and a fixing device that facilitate the positioning of the ton bag.

(5) The object of the present application is achieved through the following technical solutions: a fixing assembly includes: a connecting sleeve; a spike piece rotatably connected to the connecting sleeve; and an elastic piece, and one end of the elastic piece is connected to the connecting sleeve,

and the other end of the elastic piece is connected to the spike piece, the elastic piece at least spirally winds around and holds the outer peripheral wall of the spike piece, and the elastic piece is used to generate elastic deformation in case that the spike piece rotates relative to the connecting sleeve in a direction opposite to the spiral winding direction of the elastic piece, an included angle exists between the direction of elastic deformation and the axial extension direction of the spike piece, so as to form a spiral convex elastic structure.

(6) A fixing device includes a turnover box and the fixing assembly according to any one of the above embodiments, and the turnover box is provided with an accommodating groove and a mounting hole which are communicated with each other, and the connecting sleeve is located in the mounting hole and is fixedly connected to the turnover box. The spike piece and the elastic piece are both located in the accommodating groove, and one end of the spike piece protrudes from the outer side of the turnover box.

(7) Compared with the conventional technology, the present application at least has the following advantages: 1) in the fixing assembly of the present application, the connecting sleeve can be fixedly mounted at the mounting position. For example, the connecting sleeve can be fixedly mounted on the inner wall of the accommodating groove of the turnover box. Since the spike piece is rotatably connected with the connecting sleeve, and one end of the elastic piece is connected to the connecting sleeve, the other end of the elastic piece is connected with the spike piece, and the elastic piece spirally winds at least the outer peripheral wall of the spike piece. In case that the ton bag is placed in the accommodating groove in the turnover box and the spike piece is pierced and inserted into the surface layer of the ton bag, the spike piece rotates relative to the connecting sleeve in the direction opposite to the spiral winding direction of the elastic piece. The elastic piece is used to generate elastic deformation in case that the spike piece rotates relative to the connecting sleeve in a direction opposite to the spiral winding direction of the elastic piece. An included angle exists between the direction of elastic deformation and the axial extension direction of the spike piece, so that the elastic piece bounces outwards and forms a spiral convex elastic structure which opens outwards to the outer peripheral wall of the spike piece, which plays a role in elastic pressing and fixing the inner wall of the ton bag. In this way, it plays a fixing and limiting role similar to the "nail cap", and achieves a better fixing effect on the ton bag, that is, the ton bag is better fixed and positioned in the turnover box, so as to further dump the ton bag, realize the whole process of fixing and positioning the ton bags conveniently and quickly, and solve the problem of high positioning difficulty of the ton bags; 2) In the fixing assembly of the present application, in case that the ton bag does not need to be fixed, it is only necessary to rotate the spike piece relative to the connecting sleeve in the same direction as the spiral winding direction of the elastic piece. That is, the elastic piece can be automatically reset and rotated and tightened around the peripheral wall of the spike piece, the spiral convex elastic structure shrinks in and tightly abuts on the outer peripheral wall surface of the spike piece, that is, the elastic piece spirally winds around and holds the outer peripheral wall of the spike piece. The spike piece can be quickly pulled out from the surface of the ton bag, realizing the state that the ton bag is immediately released in case that the fixed positioning is not required, and avoiding damaging the structure of the ton bag due to the fixed positioning, and realizing the recycling of the ton bag; 3) In the fixing assembly of the present application, in case that the spike piece is pierced and inserted into the surface layer of the ton bag, the spike piece rotates in the forward and reverse directions relative to the connecting sleeve, which may realize quick switching between two different states of fixed positioning and unfixed positioning of the ton bag by the fixing assembly, the convenience of using the fixing assembly and the positioning efficiency are greatly improved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) To describe technical solutions in the embodiments of the present application more clearly, drawings required in the embodiments of the present application are introduced simply hereinafter. It should be understood that the following drawings shows some embodiments of the present application, which does not limit the scope. It is apparent that those skilled in the art can obtain other drawings based on these drawings without any creative work.
- (2) FIG. 1 is a schematic structural diagram of a use state of a fixing device of an embodiment;
- (3) FIG. 2 is a schematic structural diagram of another use state of the fixing device shown in FIG. 1;
- (4) FIG. 3 is a schematic structural diagram of a fixing assembly of the fixing device shown in FIG. 1;
- (5) FIG. 4 is a schematic structural diagram of another state of the fixing assembly shown in FIG. 3;
- (6) FIG. 5 is a schematic cross-sectional view of the fixing assembly shown in FIG. 4;
- (7) FIG. 6 is a schematic structural diagram of the elastic piece of the fixing assembly shown in FIG. 1 being turned away;
- (8) FIG. 7 is a schematic structural diagram of a fixing assembly of a fixing device according to another embodiment;
- (9) FIG. 8 is a schematic structural diagram of another state of the fixing assembly shown in FIG. 7;
- (10) FIG. 9 is a schematic cross-sectional view of the fixing assembly shown in FIG. 8;
- (11) FIG. 10 is an exploded view of a fixing assembly of a fixing device according to another embodiment;
- (12) FIG. 11 is a partial structural diagram of a fixing device according to yet another embodiment;
- (13) FIG. 12 is a schematic structural diagram of the fixing device shown in FIG. 11 in another state;
- (14) FIG. 13 is a partial enlarged schematic diagram of the fixing device shown in FIG. 11;
- (15) FIG. 14 is a cross-sectional view taken along line A-A of the fixing device shown in FIG. 11;
- (16) FIG. 15 is a schematic cross-sectional view of a fixing assembly of a fixing device according to yet another embodiment;
- (17) FIG. 16 is a schematic structural diagram of the connecting sleeve of the fixing assembly shown in FIG. 15 from another perspective;
- (18) FIG. 17 is a structural diagram of the clamping sealing ring of the fixing assembly shown in FIG. 15 from another perspective;
- (19) FIG. 18 is a schematic structural diagram of a clamping sealing ring of a fixing assembly of a fixing device according to yet another embodiment;
- (20) FIG. 19 is a partial enlarged schematic diagram of the fixing device shown in FIG. 1;
- (21) FIG. 20 is a partial structural diagram of a fixing device according to yet another embodiment;
- (22) FIG. 21 is a schematic structural diagram of a state of a fixing device according to yet another embodiment;
- (23) FIG. 22 is a schematic structural diagram of the fixing device shown in FIG. 21 in another state.

DETAILED DESCRIPTION OF THE EMBODIMENTS

- (24) In order to facilitate the understanding of the present application, the present application will be described in full details with reference to the relevant drawings. The preferred embodiments of the present application are shown in the drawings. However, the present application can be implemented in many different forms and is not limited to the embodiments described herein. On the contrary, the purpose of providing these embodiments is to make the disclosure of the present application more through and comprehensive. It should be noted that when an element is referred to

as being “fixed” to the other element, it can be directly on the other element or another element is existed between them. When an element is considered to be “connected” to the other element, it can be directly connected to the other element or another element is connected between them. As used herein, the terms “vertical”, “horizontal”, “left”, “right” and similar expressions are for the purpose of illustration only, and do not imply an only embodiment. Unless otherwise defined, all technical and scientific terms used herein have the same meanings as commonly understood by those skilled in the technical field of the present application. The terms used in the specification of the present application are only for the purpose of describing specific embodiments, and are not intended to limit the present application. The term “and/or” as used herein includes any and all combinations of one or more related listed items.

(25) As shown in FIG. 1 and FIG. 2, a fixing device **10** in an embodiment includes a turnover box **100** and a fixing assembly **200**, and the turnover box **100** is provided with an accommodating groove **102** and a mounting hole **104** which are communicated with each other. In this embodiment, the fixing device **10** is used to fix the ton bag **20** in order to carry out the pouring operation. The accommodating groove **102** is used for accommodating a ton bag. The fixing assembly **200** is connected to the turnover box **100**, and the fixing assembly **200** is partially located in the accommodating groove **102** for fixing the ton bag.

(26) Referring to FIG. 3, in one of the embodiments, the fixing assembly **200** includes a connecting sleeve **210**, a spike piece **220** and an elastic piece **230**. The connecting sleeve **210** is located in the mounting hole **104** and is fixedly connected to the turnover box **100**. The spike piece **220** is rotatably connected with the connecting sleeve **210**. One end of the elastic piece **230** is connected to the connecting sleeve **210**, and the other end of the elastic piece **230** is connected to the spike piece **220**. The elastic piece **230** at least spirally winds around and is secured on the outer peripheral wall of the spike piece **220**. That is, the elastic piece **230** at least spirally winds around and holds the outer peripheral wall of the spike piece **220** along the winding direction. It can be understood that, in one of the embodiments, the elastic piece **230** is connected to the spike piece **220** and the connecting sleeve **210**, and in case that the elastic piece **230** is not deformed, the elastic piece **230** at least spirally winds around and holds the outer peripheral wall of the spike piece **220**, that is, the elastic piece **230** at least spirally tightly winds around and holds the outer peripheral wall of the spike piece **220**.

(27) As shown in FIG. 4 and FIG. 5, the elastic piece **230** is used to generate elastic deformation in case that the spike piece **220** rotates relative to the connecting sleeve **210** in a direction opposite to the spiral winding direction of the elastic piece **230**, and there is an included angle between the direction of elastic deformation and the axial extension direction of the spike piece **220**. That is, the direction of elastic deformation is not parallel to the axial extension direction of the spike piece **220** to form a spiral convex elastic structure, so that the spiral convex elastic structure elastically compresses and fixes the inner wall of the ton bag. That is, the spiral convex elastic structure generates an axial force parallel to the axial direction of the spike piece **220** on the inner wall of the ton bag, thus playing a role similar to the “nail cap” for elastically fixing and limiting the ton bag, thus achieving a better fixing effect on the ton bag. In this embodiment, the so-called spiral convex elastic structure is a structural shape in which the elastic piece **230** is elastically deformed as the spike piece **220** rotates relative to the connecting sleeve **210** in a direction opposite to the spin direction of the elastic piece **230**. The elastic piece **230** that generates elastic deformation spirally winds the spike piece **220** and is in a state of elastically expanding outwards. As shown in FIG. 4, the elastic piece **230** that generates elastic deformation does not completely p the outer peripheral wall of the spike piece **220**. That is, the elastic piece **230** that generates elastic deformation is partially separated from the outer peripheral wall of the spike piece **220**, and the elastic piece **230** is rotated and unfolded, so that the spiral convex elastic structure elastically presses and fixes the inner wall of the ton bag. It can be understood that while the spike piece **220** rotates relative to the connecting sleeve **210** in the same direction as the winding direction of the elastic piece **230**, the

elastic piece gradually resets. The spiral convex elastic structure gradually disappears until the elastic piece spirally winds around and holds the outer peripheral wall of the spike piece, so that the elastic piece is tightly attached to the outer peripheral wall of the spike piece.

(28) As shown in FIG. 1 and FIG. 3, in this embodiment, the spike piece **220** and the elastic piece **230** are both located in the accommodating groove **102**. Moreover, one end of the spike piece **220** protrudes from the outer side of the turnover box **100**, so that the spike piece **220** may be rotated relative to the connecting sleeve **210** on the outer side of the turnover box **100**, which improves the convenience of using the fixing assembly **200**. In one embodiment, the spike piece **220** is located at the bottom of the accommodating groove **102**, so that the spike piece **220** may pierce and position the bottom of the ton bag. Due to the gravity of the ton bag, the spike piece **220** may better pierce and position the bottom of the ton bag.

(29) In the above fixing device **10** and fixing assembly **200**, the connecting sleeve **210** can be fixedly mounted at the mounting position. For example, the connecting sleeve **210** may be fixedly mounted on the inner wall of the accommodating groove of the turnover box **100**. Since the spike piece **220** is rotatably connected with the connecting sleeve **210**, and one end of the elastic piece **230** is connected to the connecting sleeve **210**, the other end of the elastic piece **230** is connected with the spike piece **220**, and the elastic piece **230** at least spirally winds on the outer peripheral wall of the spike piece **220**. In case that the ton bag is placed in the accommodating groove in the turnover box **100** and the spike piece **220** is pierced and inserted into the surface layer of the ton bag, the spike piece **220** rotates relative to the connecting sleeve **210** in the direction opposite to the spiral winding direction of the elastic piece **230**. The elastic piece **230** is used to generate elastic deformation in case that the spike piece **220** rotates relative to the connecting sleeve **210** in a direction opposite to the spiral winding direction of the elastic piece **230**. An included angle exists between the direction of elastic deformation and the axial extension direction of the spike piece **220**, so that the elastic piece **230** bounces outwards and forms a spiral convex elastic structure which opens outwards to the outer peripheral wall of the spike piece **220**, which plays a role in elastic pressing and fixing the inner wall of the ton bag. In this way, it plays a fixing and limiting role similar to the “nail cap”, and achieves a better fixing effect on the ton bag, that is, the ton bag is better fixed and positioned in the turnover box **100**, so as to further perform the dumping of the ton bag, realize the whole process of fixing and positioning the ton bags conveniently and quickly, and solve the problem of high positioning difficulty of the ton bags; in case that the ton bag does not need to be fixed, it is only necessary to rotate the spike piece **220** relative to the connecting sleeve **210** in the same direction as the spiral winding direction of the elastic piece **230**. That is, the elastic piece **230** can be automatically reset and rotated and tightened around the peripheral wall of the spike piece **220**, the spiral convex elastic structure shrinks in and tightly holds the outer peripheral wall surface of the spike piece **220**, that is, the elastic piece **230** spirally winds around and holds the outer peripheral wall of the spike piece **220**. The spike piece **220** can be quickly pulled out from the surface layer of the ton bag, realizing the state that the ton bag is immediately released in case that the fixed positioning is not required, and avoiding damaging the structure of the ton bag due to the fixed positioning, and realizing the recycling of the ton bag; in the fixing assembly **200** of the present application, in case that the spike piece **220** is pierced and inserted into the surface layer of the ton bag, the spike piece **220** rotates in the forward and reverse directions relative to the connecting sleeve **210**, which may realize quick switching between two different states of fixed positioning and unfixed positioning of the ton bag by the fixing assembly **200**, the convenience of using the fixing assembly **200** and the positioning efficiency are greatly improved.

(30) As shown in FIG. 4, in one embodiment, a part of the elastic piece **230** spirally presses the outer peripheral wall of the spike piece **220**, and another part of the elastic piece **230** spirally presses the connecting sleeve **210**, so that the elastic piece **230** has a circumferential pressing length on the surface of the connecting sleeve **210** in the vicinity of the end connected to the connecting sleeve **210**. Under the condition of the same spiral circumference size, in case that the

spike piece **220** rotates relative to the connecting sleeve **210**, the fixing assembly **200** can be switched to quickly fix or unfix the ton bag. That is, the spiral convex elastic structure can be formed or eliminated, that is, the elastic deformation of the elastic member **230** can be quickly generated or eliminated. It can be understood that, in other embodiments, the elastic piece **230** may also be completely spirally wound and pressed on the outer peripheral wall of the spike piece **220**, and one end of the elastic piece **230** is connected to the edge of the connecting sleeve **210**.

(31) In one of the embodiments, the direction of elastic deformation and the axial extension direction of the spike piece **220** are perpendicular to each other. That is, the direction of elastic deformation is parallel to the plane where the radial direction of the spike piece **220** is located. In case that the spike piece **220** rotates relative to the connecting sleeve **210** in a direction opposite to the spiral winding direction of the elastic piece **230**, the elastic piece **230** is elastically deformed in the radial direction of the spike piece **220**. In turn, the spiral convex elastic structure may better elastically press and fix the inner wall of the ton bag. It can be understood that in other embodiments, the direction of elastic deformation is not limited to being perpendicular to the axial extension direction of the spike piece **220**. For example, the included angle between the direction of elastic deformation and the axial extension direction of the spike piece **220** is greater than 5° and less than or equal to 80° .

(32) As shown in FIG. 4, in one of the embodiments, the elastic piece **230** is a spring wire structure, so that the elastic piece **230** has better elastic strength, and the deformed spiral convex elastic structure may better elastically press and fix the inner wall of the ton bag. It can be understood that in other embodiments, the elastic piece **230** is not limited to a spring wire structure, and may also be a coil spring or an elastic rubber piece or other elastic pieces **230**.

(33) As shown in FIG. 6, in one embodiment, the cross-sectional diameter of the elastic piece **230** decreases from the middle of the elastic piece **230** to both ends of the elastic piece **230**, so that the elastic deformation force generated by the elastic piece **230** can better press and act on the ton bag. The spiral convex elastic structure can more firmly hold the inner wall of the ton bag, and the deformed spiral convex elastic structure can better elastically press and fix the inner wall of the ton bag.

(34) As shown in FIG. 5, in one of the embodiments, the spike piece **220** includes a spike end **222**, a winding portion **224**, and a rotating connecting portion **226** that are connected in sequence. The rotating connecting portion **226** is rotatably connected with the connecting sleeve **210**, one end of the elastic piece **230** is connected to the winding portion **224**, and the elastic piece **230** spirally winds around and holds the outer peripheral wall of the winding portion **224**, so that the elastic piece **230** is connected to the spike piece **220**, the elastic piece **230** spirally winds around and holds the spike piece **220**, and the spike piece **220** is rotatably connected to the connecting sleeve **210**. In this embodiment, the spike end **222** is used to pierce the surface layer of the ton bag. Further, the spike end **222** has a conical structure, so that the spike end **222** can better pierce the surface layer of the ton bag.

(35) In order to allow the spike piece **220** to quickly pierce into the ton bag, as shown in FIG. 1, in this embodiment, the spike piece **220** is provided at the bottom of the inner wall of the accommodating groove of the turnover box **100**, so that the spike piece **220** can pierce the surface layer inserted into the bottom of the ton bag in case that the ton bag is placed in the turnover box **100**. In this way, under the gravity of the ton bag, the spike piece **220** can quickly pierce and be inserted into the ton bag.

(36) As shown in FIG. 7 to FIG. 9, in one of the embodiments, a limit step **224a** is formed on the outer peripheral wall of the end, adjacent to the spike end **222**, of the winding portion **224**. That is, the diameter of the winding portion **224** for spirally winding the outer peripheral wall of the elastic piece **230** is smaller than the diameter of the adjacent winding portion **224** of the spike end **222**. An accommodating groove **229** for accommodating the elastic piece **230** is formed on the outer peripheral wall of the winding portion **224** of the spike piece **220**. One end of the elastic piece **230**

is connected to the surface of the limit step **224a**, and the elastic piece **230** spirally winds around and holds and is limited to the limit step **224a**. That is, the elastic piece **230** winds around and holds the surface of the limit step **224a**, so that the elastic piece **230** reliably winds around and holds the winding portion **224**, and the elastic piece **230** is reliably confined to one side of the limit step **224a**. The elastic piece **230** is prevented from being separated from the spike piece **220** in the process of pulling out the ton bag with the spike piece **220**, and the cross section of the spike piece **220** is reduced, and that the elastic piece **230** protrudes from the outer peripheral wall of the spike piece **220** and causes greater resistance to the spike piece **220** to pierce or pull out the ton bag is also avoided, thereby the resistance of the fixing assembly **200** piercing or pulling out the ton bag is reduced, which is convenient for the fixing assembly **200** to quickly pierce or pull out the ton bag. It can be understood that when the spike piece **220** does not rotate relative to the connecting sleeve **210** in a direction opposite to the spinning direction of the elastic piece **230**, the elastic piece **230** is in a spiral shape and tightly winds around and holds the outer peripheral wall of the winding portion **224**. In case that the spike piece **220** rotates in a direction opposite to the spinning direction of the elastic piece **230** relative to the connecting sleeve **210**, the elastic piece **230** is elastically deformed. At this time, although the elastic piece **230** that generates elastic deformation still spirally winds the spike piece **220**, the elastic piece **230** is in a state of elastically expanding outward. That is, the elastic piece **230** that generates elastic deformation does not tightly hold the outer peripheral wall of the spike piece **220**, so that the spiral convex elastic structure elastically presses and fixes the inner wall of the ton bag.

(37) As shown in FIG. 7 to FIG. 9, further, the diameter of the rotating connecting portion **226** is smaller than the diameter of the winding portion **224**, so that the two ends of the elastic piece **230** are respectively connected to the connecting sleeve **210** and the spike piece **220**, and the elastic piece **230** can more closely wind the outer peripheral wall of the spike piece **220**. That is, the elastic piece **230** can wind and hold the peripheral wall of the spike piece **220** smoothly and tightly, so that the fixing assembly **200** can better pierce the ton bag and fix the ton bag. In this embodiment, the diameter of the winding portion **224** is equal to the diameter of the connecting sleeve **210**, so that the elastic piece **230** spirally winds the outer peripheral wall of the spike piece **220** more smoothly. The spiral winding structure of the elastic piece **230** at the connection between the winding portion **224** and the connecting sleeve **210** is made smoother.

(38) As shown in FIG. 10, further, a first thread **226a** is formed on the outer peripheral wall of the rotating connecting portion **226**, and a second thread **211** is formed on the inner peripheral wall of the connecting sleeve **210**. The connecting sleeve **210** is mounted outside the rotating connecting portion **226** and is threadedly connected to the rotating connecting portion **226**, so that the connecting sleeve **210** is rotationally connected to the rotating connecting portion **226**. In addition, the rotating connecting portion **226** makes the spike piece **220** move relative to the connecting sleeve **210** in the axial direction of the spike piece **220** during the relative rotation of the connecting sleeve **210**. The part where the elastic piece **230** is connected to the spike piece **220** is a first end, and the part where the elastic piece **230** and the connecting sleeve **210** are connected is a second end. In case that the spike piece **220** rotates in a direction opposite to the winding direction of the elastic piece **230** relative to the connecting sleeve **210**, the spike piece **220** and the connecting sleeve **210** move relatively, reducing the distance between the first end and the second end. The spiral convex elastic structure formed in this way is flatter, which improves the axial acting force of the spiral convex elastic structure on the inner wall of the ton bag, and further enables the spiral convex elastic structure to better press and fix the inner wall of the ton bag. In this embodiment, the first thread **226a** and the second thread **211** are mutually adapted threaded connection structures, so that the rotating connecting portion **226** and the connecting sleeve **210** are threadedly connected. Further, the first thread **226a** spirally winds the outer peripheral wall of the rotating connecting portion **226**, and the second thread **211** spirally winds the inner peripheral wall of the connecting sleeve **210**, so that the first thread **226a** and the second thread **211** are better threadedly connected,

and the adjustment of the axial connection position between the spike piece **220** and the connecting sleeve **210** is more flexible. That is, the adjustment of the distance between the first end and the second end is more flexible.

(39) As shown in FIG. **9**, in one embodiment, the connecting sleeve **210** is mounted outside the spike piece **220** and rotationally connected with the spike piece **220**, so that the relative rotational connection between the connecting sleeve **210** and the spike piece **220** is more stable. In this embodiment, the connecting sleeve **210** is mounted outside the outer peripheral wall of the rotating connecting portion **226** and is rotatably connected with the rotating connecting portion **226**. Further, the first thread **226a** is spirally arranged on the outer peripheral wall of the rotating connecting portion **226**, the second thread **211** is spirally arranged on the inner peripheral wall of the connecting sleeve **210**, so that the connecting sleeve **210** and the spike piece **220** are threadedly connected.

(40) In order to make the structure of the fixing assembly **200** relatively simple, as shown in FIG. **8**, in one of the embodiments, two ends of the elastic piece **230** are respectively welded to the outer peripheral wall of the spike piece **220** and the outer peripheral wall of the connecting sleeve **210**, which enables the two ends of the elastic piece **230** to be connected to the spike piece **220** and the connecting sleeve **210**, and the structure of the fixing assembly **200** is relatively simple.

(41) Due to the heavy weight of the ton bag, however, in the process of fixing the ton bag repeatedly by the fixing device **10**, both ends of the elastic piece **230** are easily broken or even separated from the spike piece **220** and the connecting sleeve **210** under the action of gravity. As a result, the fixing assembly **200** is scrapped after multiple uses, and the service life of the fixing assembly **200** is short; in addition, under the action of gravity, the elastic piece **230** of the fixing assembly **200** is prone to plastic deformation. That is, the elastic piece **230** cannot spirally tightly wind and hold the outer peripheral wall of the spike piece **220** after resetting. That is, the elastic piece **230** locally protrudes from the outer peripheral wall of the spike piece **220**, so that the resistance of the spike piece **220** in the process of piercing the ton bag is relatively large, and the time required for the fixing assembly **200** to fix and position the ton bag is relatively long. After repeated use, the plastic deformation of the elastic piece **230** becomes larger, the fixing assembly **200** is eventually scrapped and unable to be used further, thus shortening the service life of the fixing assembly **200**. In order to improve the service life and convenience of using the fixing assembly **200**, as shown in FIG. **10**, in another embodiment, two ends of the elastic piece **230** are detachably connected to the outer peripheral wall of the spike piece **220** and the outer peripheral wall of the connecting sleeve **210**, so as to perform regular maintenance or replacement of the elastic piece **230**, and the service life and convenience of using the fixing assembly **200** are improved. In one of the embodiments, the first end of the elastic piece **230** is provided with a first bending portion **230a**, and the outer peripheral wall of the spike piece **220** is provided with a first insertion hole **223**, The first bending portion is inserted into the first insertion hole **223**, so that the elastic piece **230** is detachably connected to the outer peripheral wall of the spike piece **220**. The second end of the elastic piece **230** is provided with a second bending portion **230b**, and the outer peripheral wall of the connecting sleeve **210** is provided with a second insertion hole **213**, the second bending portion is inserted into the second insertion hole **213**, so that the elastic piece **230** is detachably connected to the outer peripheral wall of the connecting sleeve **210**. In order to avoid the problem that the opening of the second insertion hole **213** affects the structural strength of the connecting sleeve **210**, further, a mounting boss **212** is protrudingly provided on the outer peripheral wall of the connecting sleeve **210**, the second insertion hole **213** is formed in the mounting boss **212**, so that the position where the second insertion hole **213** is formed in the connecting sleeve **210** has better structural strength. The problem that the opening of the second insertion hole **213** affects the structural strength of the connecting sleeve **210** is avoided.

(42) It can be understood that, as shown in FIG. **11** and FIG. **12**, in other embodiments, the two ends of the elastic piece **230** are not limited to be connected to the connecting sleeve **210** and the

spike piece **220** in a plug-in manner. In one of the embodiments, the first end of the elastic piece **230** and the second end of the elastic piece **230** are both parallel to the cross section of the spike piece **220**. Referring to FIG. **13** and FIG. **14**, the first end of the elastic piece **230** is formed with a first curved hook **230c**, and the first curved hook **230c** is formed with a first fixing hole **233**. A second curved hook **230d** is formed at the second end of the elastic piece **230**, and a second fixing hole **235** is formed in the second curved hook **230d**. The fixing assembly **200** further includes a first locking piece **240**, a first connecting hole **225** is formed in the outer peripheral wall of the spike piece **220**, and the first locking piece **240** is inserted into the first fixing hole **233** and the first connecting hole **225**. The first locking piece **240** fixedly connects the first end to the spike piece **220**, thereby realizing the detachable connection between the elastic piece **230** and the spike piece **220**. The fixing assembly **200** further includes a second locking piece **250**, and a second connecting hole **215** is defined on the outer peripheral wall of the connecting sleeve **210**. The second locking piece **250** is inserted into the second fixing hole **235** and the second connecting hole **215**, so that the second locking piece **250** fixedly connects the second end to the connecting sleeve **210**, thereby realizing the detachable connection between the elastic piece **230** and the connecting sleeve **210**. In this embodiment, the cross section of the spike piece **220** is perpendicular to the axial direction of the spike piece **220**. The first connecting hole **225** and the second connecting hole **215** may be threaded holes, and the first locking piece **240** and the second locking piece **250** may be screws or bolts, so that the structures of the first locking piece **240** and the second locking piece **250** are relatively simple.

(43) In order to make the first locking piece **240** better fix the first end to the spike piece **220**, and make the second locking piece **240** better fix the second end to the connecting sleeve **210**, as shown in FIG. **13**, further, a surface, facing away from the spike piece **220**, of the first curved hook **230c** is provided with a first abutting inclined surface **237**, a first pressing inclined surface **241** is provided at a position where the first locking piece **240** abuts the first curved hook **230c**, the first pressing inclined surface **241** abuts against the first abutting inclined surface **237**, and prevents the first locking piece **240** from shaking due to the poor abutment reliability at the abutting position with the first end, and the first locking piece **240** can more firmly connect the first end to the spike piece **220**. A surface, facing away from the connecting sleeve **210**, of the second curved hook **230d** is provided with a second abutting inclined surface **239**, a second pressing inclined surface **251** is provided at a position where the second locking piece **250** abuts the second curved hook **230d**, the second pressing inclined surface **251** abuts against the second abutting inclined surface **239**, and prevents the second locking piece **250** from shaking due to the poor abutment reliability at the abutting position with the second end, and the second locking piece **250** can more firmly connect the second end to the connecting sleeve **210**.

(44) In order to make the first locking piece **240** better abut against and press the first end, the fixing assembly **200** further includes a first elastic pressing ring. The first elastic pressing ring is mounted outside the first locking piece **240**, and the first elastic pressing ring elastically abuts against the inner peripheral wall of the first curved hook **230c**, so that the first locking piece **240** can better abut against and press the first end. In order to make the second locking piece **250** abut and press against the second end better, the fixing assembly **200** further includes a second elastic pressing ring. The second elastic pressing ring is sheathed on the second locking piece **250**, and the second elastic pressing ring elastically abuts against the inner peripheral wall of the second curved hook **230c**, so that the second locking piece **250** can better abut against and press the second end. In this embodiment, the first elastic pressing ring and the second elastic pressing ring are both elastic silicone rings.

(45) In order to avoid the interference of the first locking piece **240** and the second locking piece **250** in the process of fixing and positioning the ton bag, which causes the problem of greater resistance for the fixing assembly **200** to pierce the ton bag, as shown in FIG. **13** and FIG. **14**, further, the first connecting hole **225** is defined at the end face, adjacent to the winding portion **224**,

of the spike end **222**. That is, the first connecting hole **225** is formed in the inner peripheral wall of the limit step **224a**, so that the first locking piece **240** is accommodated in the accommodating groove **229** in case that the first end is fixedly connected to the spike piece **220**. The second connecting hole **215** is formed in the end face of the connecting sleeve **210** near the spike end **222**, so that the second locking piece **250** is accommodated in the accommodating groove **229** in case that the second end is fixedly connected to the connecting sleeve **210**, and avoid the problem that the first locking piece **240** and the second locking piece **250** interfere in the process of fixing and positioning the ton bag, which causes greater resistance for the fixing assembly **200** to pierce the ton bag. In this embodiment, the diameter of the spike end **222** is greater than the diameter of the winding portion **224**, so that the connection between the spike end **222** and the winding portion **224** forms a first connecting end surface, and the first connecting hole **225** is defined on the first connecting end surface. The diameter of the rotating connecting portion **226** is equal to the diameter of the winding portion **224**, and the diameter of the spike end **222** is equal to the outer diameter of the connecting sleeve **210**, the connecting sleeve **210** is mounted outside and rotationally connected to the rotating connecting portion **226**. Since the diameter of the spike end **222** is greater than the diameter of the winding portion **224**, the side, adjacent to the spike end, of the connecting sleeve **210** forms a second connecting end surface, the second connecting hole **215** is defined at the second connecting end face, so that an accommodating groove **229** is formed between the spike end **222** and the connecting sleeve **210**, and then the elastic piece **230**, the first locking piece **240** and the second locking piece **250** are all accommodated in the accommodating groove **229**. The first end of the elastic piece **230** is fixed to the first connecting end face by the first locking piece **240**, and the second end of the elastic piece **230** is fixed to the second connecting end face by the second locking piece **250**.

(46) As the fixing device **10** is used for fixing and positioning ton bags, there is more dust in the surrounding environment of the fixing device **10**. Therefore, it is inevitable that part of dust enters through the joint between the spike piece **220** and the connecting sleeve **210**, resulting in serious wear at the connection between the spike piece **220** and the connecting sleeve **210**. In order to avoid the problem of severe abrasion at the connection between the spike piece **220** and the connecting sleeve **210**, as shown in FIG. **15** to FIG. **17**, further, a positioning ring groove **216** is formed at the end of the connecting sleeve **210** adjacent to the spike end **222**, and the positioning ring groove **216** is disposed around the spike end **222**. The fixing assembly **200** further includes a clamping sealing ring **260**, the clamping sealing ring **260** is movably sheathed on the spike piece **220**, and the clamping sealing ring **260** is arranged in the positioning ring groove **216** and elastically abuts against the connecting sleeve **210**, which improves the sealing performance of the connection between the spike piece **220** and the connecting sleeve **210**, and avoids the problem of serious wear at the connection between the spike piece **220** and the connecting sleeve **210**, thereby prolonging the service life of the fixing assembly **200**.

(47) It can be understood that since the connection between the spike piece **220** and the connecting sleeve **210** requires repeated relative movements, such as the relative rotation of the spike piece **220** and the connecting sleeve **210**, as the service time or frequency increases, the clamping sealing ring **260** is severely worn and cannot achieve a better sealing effect. As shown in FIG. **15** to FIG. **17**, further, the clamping sealing ring **260** is detachably connected to the spike piece **220**, so that the clamping sealing ring **260** is replaced regularly, thereby avoiding the problem that the clamping sealing ring **260** is seriously worn and cannot play a better sealing effect. Furthermore, the clamping sealing ring **260** is provided with a stop protrusion **261**, and the inner peripheral wall of the positioning ring groove **216** is provided with a rotation limit stop **216a**. There is a clamping gap between the rotation limit stop **216a** and the bottom of the positioning ring groove **216**. The stop protrusion is used to screw into the clamping gap in case of rotating at a predetermined angle in the positioning ring groove **216**. The stop protrusion abuts against the rotation limit stop **216a**, so that the clamping sealing ring **260** is clamped and limited in the positioning ring groove **216**, and the

clamping sealing ring **260** is detachably connected to the spike piece **220**. In this embodiment, the clamping sealing ring **260** may be a silicone ring or a rubber ring. In order to make the clamping sealing ring **260** more tightly and elastically abut in the positioning ring groove **216**, and referring to FIG. **18**, further, the outer peripheral wall of the clamping sealing ring **260** is provided with a sealing flange **263**. The sealing flange is located in the positioning ring groove **216** and elastically abuts against the connecting sleeve **210**, so that the clamping sealing ring **260** elastically abuts against the positioning ring groove **216** more tightly. In this embodiment, the cross section of the sealing flange is triangular or rectangular or ladder-shaped, the sealing flange is located in the positioning ring groove **216** and is tightly connected with the inner peripheral wall of the connecting sleeve **210**, so as to better avoid the problem of dust entering the connection between the connecting sleeve and the spike piece.

(48) In order to make the spike piece **220** rotate more smoothly relative to the connecting sleeve **210**, as shown in FIG. **1** and FIG. **19**, further, the fixing device **10** further includes a support bearing **300**, the support bearing **300** is mounted outside the spike piece **220**, and the support bearing **300** is fixedly mounted on the turnover box **100**, the end, protruding from the outer side of the turnover box **100**, of the spike piece **220** is more stable in the process of rotating relative to the connecting sleeve **210**, and the spike piece **220** rotates more smoothly relative to the connecting sleeve **210**. However, since the fixing device **10** is used to fix and position the ton bag, there is more dust in the surrounding environment of the fixing device **10**, and there is inevitably a problem that some dust enters the support bearing **300** and the support bearing **300** is easily worn out. As shown in FIG. **20**, further, the fixing device **10** further includes a dust cover **400**, the dust cover **400** covers the support bearing **300**, and the dust cover **400** is provided with a first avoiding hole **402** and a second avoiding hole **404**. The spike piece **220** penetrates through the first avoiding hole **402** and the second avoiding hole **404**. In this embodiment, a first sealing ring **402a** is protrudingly provided on the inner peripheral wall of the first avoiding hole **402**, and a second sealing ring **404a** is protrudingly provided on the inner peripheral wall of the second avoiding hole **404**. The first sealing ring **402a** and the second sealing ring **404a** are mounted outside the spike piece **220**, so that the support bearing **300** is sealed in the dust cover **400**. Furthermore, the dust in the surrounding environment of the fixing device **10** is not easy to enter the dust cover **400**, and the service life of the support bearing **300** is improved. Furthermore, an oil injection hole **406** is defined on the side wall of the dust cover **400**, the oil injection hole **406** is arranged corresponding to the balls of the support bearing **300**, so that a user may regularly maintain the support bearing **300** through the oil injection hole **406**, and the service life of the support bearing **300** is prolonged.

(49) In order to improve the reliability of the fixing and positioning of the ton bag, as shown in FIG. **1**, in one of the embodiments, the number of the fixing assembly **200** is multiple, the multiple fixing assemblies **200** are spaced apart, so that the ton bag can be fixed and positioned through the multiple fixing assemblies **200**, and the reliability of fixing and positioning the ton bag is improved. In this embodiment, multiple fixing assemblies **200** are arranged at the bottom of the turnover box **100** and spaced apart, so that the fixing assemblies **200** fix and position the bottom of the ton bag.

(50) As shown in FIG. **1**, in one of the embodiments, the end, protruding from the outer side of the turnover box **100**, of the spike piece **220** is provided with a wrench fitting portion **221**, during operation, the wrench tool may act on the wrench fitting portion **221** to make the spike piece **220** rotate relative to the connecting sleeve **210**, thus improving the convenience of using the fixing assembly **200**. In this embodiment, the wrench fitting portion **221** is a wrench fixing buckle with a square section. Referring to FIG. **2**, the fixing device **10** further includes a fixing frame **600**. The fixing frame **600** is fixed and supported on the turnover box **100**, and a hollow cavity **601** is formed on the side wall of the fixing frame **600**. The wrench fitting portion **221** is located in the hollow cavity, and a wrench tool is used to act on the wrench fitting portion **221** through the hollow cavity.

(51) It can be understood that in other embodiments, the end, protruding outside the turnover box

100, of the spike piece **220** is not limited to be rotated by setting the wrench fitting portion **221**, and the elastic piece **230** is elastically deformed to form a spiral convex elastic structure, so that the spiral convex elastic structure elastically presses and fixes the inner wall of the ton bag. As shown in FIG. **21** and FIG. **22**, in one of the embodiments, the fixing device **10** further includes a fixing frame **600** and a driving mechanism **700**, the fixing frame **600** is fixed and supported on the turnover box **100**, and the driving mechanism **700** is mounted on the fixing frame **600**. The power output end of the driving mechanism **700** is connected to the end of the spike piece **220** protruding from the outer side of the turnover box **100**, so that the driving mechanism **700** drives the spike piece **220** to rotate relative to the connecting sleeve **210**, the elastic piece **230** is elastically deformed to form a spiral convex elastic structure, so that the spiral convex elastic structure elastically presses and fixes the inner wall of the ton bag. In this embodiment, the driving mechanism **700** may be a motor driving mechanism.

(52) As shown in FIG. **21** and FIG. **22**, in one of the embodiments, the number of the fixing assembly **200** is multiple, the multiple fixing assemblies **200** are arranged side by side on the turnover box **100**. The fixing device **10** further includes multiple transmission gears **800**, the multiple transmission gears **800** are sheathed on the spike pieces **220** of the multiple fixing assemblies **200** in a one-to-one correspondence, and the transmission gears **800** are all located outside the turnover box **100**. The transmission gears **800** are sequentially meshed and transmitted along the arrangement direction of the fixing assemblies **200**, and the spike piece **220** of one of the fixing assemblies **200** is connected to the power output end of the driving mechanism **700**. In case that the driving mechanism **700** drives the spike piece **220** of one of the fixing assemblies **200** to rotate relative to the corresponding connecting sleeve **210**, the transmission gear **800** corresponding to the fixing assembly **200** transmits the power of the driving mechanism **700** to other transmission gears **800**. In this way, the spike pieces **220** of the multiple fixing assemblies **200** rotate relative to the corresponding connecting sleeve **210** at the same time, and multiple fixing assemblies **200** simultaneously fix and position or release the fixed positioning of the ton bag in the turnover box **100**, which further improves the convenience of using the fixing device **10**. In this embodiment, the elastic pieces **230** of two adjacent fixing assemblies **200** spirally winding and abutting against the outer peripheral wall of the corresponding spike piece **220** in opposite spiral directions. In this way, in case that the driving mechanism **700** drives the spike pieces **220** of one of the fixing assemblies **200** to rotate relative to the corresponding connecting sleeve **210**, the rotation directions of the spike pieces **220** of two adjacent fixing assemblies **200** are opposite, and the elastic pieces **230** of the multiple fixing assemblies **200** may simultaneously generate elastic deformation or release elastic deformation. In turn, the multiple fixing assemblies **200** may simultaneously fix or unfix the ton bags in the turnover box **100**.

(53) Compared with the conventional technology, the present application at least has the following advantages: 1. In the fixing assembly **200** of the present application, the connecting sleeve **210** can be fixedly mounted at the mounting position. For example, the connecting sleeve **210** can be fixedly mounted on the inner wall of the accommodating groove of the turnover box **100**. Since the spike piece **220** is rotatably connected with the connecting sleeve **210**, and one end of the elastic piece **230** is connected to the connecting sleeve **210**, the other end of the elastic piece **230** is connected with the spike piece **220**, and the elastic piece **230** spirally winds at least the outer peripheral wall of the spike piece **220**. In case that the ton bag is placed in the accommodating groove in the turnover box **100** and the spike piece **220** is pierced and inserted into the surface layer of the ton bag, the spike piece **220** rotates relative to the connecting sleeve **210** in the direction opposite to the spiral winding direction of the elastic piece **230**. The elastic piece **230** is used to generate elastic deformation in case that the spike piece **220** rotates relative to the connecting sleeve **210** in a direction opposite to the spiral winding direction of the elastic piece **230**. An included angle exists between the direction of elastic deformation and the axial extension direction of the spike piece **220**, so that the elastic piece **230** bounces outwards and forms a spiral convex

elastic structure which opens outwards to the outer peripheral wall of the spike piece **220**, which plays a role in elastic pressing and fixing the inner wall of the ton bag. In this way, it plays a fixing and limiting role similar to the “nail cap”, and achieves a better fixing effect on the ton bag, that is, the ton bag is better fixed and positioned in the turnover box **100**, so as to further dump the ton bag, realize the whole process of fixing and positioning the ton bags conveniently and quickly, and solve the problem of high positioning difficulty of the ton bags; 2. In the fixing assembly **200** of the present application, in case that the ton bag does not need to be fixed, it is only necessary to rotate the spike piece **220** relative to the connecting sleeve **210** in the same direction as the spiral winding direction of the elastic piece **230**. That is, the elastic piece **230** can be automatically reset and rotated and tightened around the peripheral wall of the spike piece **220**, the spiral convex elastic structure shrinks in and tightly abuts on the outer peripheral wall surface of the spike piece **220**, that is, the elastic piece **230** spirally winds around and holds the outer peripheral wall of the spike piece **220**. The spike piece **220** can be quickly pulled out from the surface of the ton bag, realizing the state that the ton bag is immediately released in case that the fixed positioning is not required, and avoiding damaging the structure of the ton bag due to the fixed positioning, and realizing the recycling of the ton bag; 3. In the fixing assembly **200** of the present application, in case that the spike piece **220** is pierced and inserted into the surface layer of the ton bag, the spike piece **220** rotates in the forward and reverse directions relative to the connecting sleeve **210**, which may realize quick switching between two different states of fixed positioning and unfixed positioning of the ton bag by the fixing assembly **200**, the convenience of using the fixing assembly and the positioning efficiency are greatly improved.

(54) The above embodiments only express several embodiments of the present application, and their descriptions are specific and detailed, but they cannot be understood as limiting the scope of the present application. It should be noted that for those skilled in the art, without departing from the concept of the present application, several modifications and improvements can be made, and these all fall within the protection scope of the present application. Therefore, the protection scope of the present application should be determined by the appended claims.

Claims

1. A fixing device, comprising a turnover box and a fixing assembly, the fixing assembly comprising: a connecting sleeve; a spike piece rotatably connected to the connecting sleeve; and an elastic piece, wherein one end of the elastic piece is connected to the connecting sleeve, and the other end of the elastic piece is connected to the spike piece, the elastic piece at least spirally winds around and holds an outer peripheral wall of the spike piece, in case that the spike piece rotates relative to the connecting sleeve in a direction opposite to the spiral winding direction of the elastic piece, the elastic piece generates elastic deformation which has an included angle with the axial extension direction of the spike piece to form a spiral convex elastic structure; wherein the turnover box is provided with an accommodating groove and a mounting hole which are communicated; the connecting sleeve is located in the mounting hole and is fixedly connected to the turnover box, the spike piece and the elastic piece are both located in the accommodating groove, and one end of the spike piece protrudes from an outer side of the turnover box.
2. The fixing device according to claim 1, wherein a direction of the elastic deformation is perpendicular to an axial extension direction of the spike piece.
3. The fixing device according to claim 1, wherein the elastic piece is a spring wire structure.
4. The fixing device according to claim 3, wherein a cross-sectional diameter of the elastic piece decreases from the middle of the elastic piece to the two ends of the elastic piece.
5. The fixing device according to claim 1, wherein the spike piece comprises a spike end, a winding portion and a rotating connecting portion that are connected in sequence, the rotating connecting portion is rotatably connected with the connecting sleeve, one end of the elastic piece is connected

to the winding portion, and the elastic piece spirally winds around and holds the outer peripheral wall of the winding portion.

6. The fixing device according to claim 5, wherein a limit step is formed on the outer peripheral wall of the end of the winding portion adjacent to the spike end, one end of the elastic piece is connected to the limit step, and the elastic piece spirally winds around and holds and is limited to the limit step.

7. The fixing device according to claim 1, wherein the connecting sleeve is mounted outside the spike piece and is rotatably connected with the spike piece.

8. The fixing device according to claim 1, wherein the end of the spike piece protruding from the outer side of the turnover box is provided with a wrench matching portion.

9. The fixing device according to claim 1, wherein the fixing device further comprises a fixing frame and a driving mechanism, the fixing frame is fixed and supported on the turnover box, and the driving mechanism is mounted on the fixing frame, and the power output end of the driving mechanism is connected with the end, protruding from the outer side of the turnover box, of the spike piece.
